

Midterm Exam

Microeconomic Theory III – ECON 709, Fall 2022

Do not open until instructed. Closed-book exam.

The exam consists of 2 questions with subquestions each. In total, you can reach 60 points; that is, 30 points for each question. The exam is from 8.35–9.55am. Good luck!

1. (30 points) *Preferences* (after Rubinstein, C5)

Identify a professor's lifetime with the interval $[0, 1]$. There are $K + 1$ academic ranks, $0, \dots, K$. All professors start at rank 0 and eventually reach rank K . Define a career as a sequence $t = (t_1, \dots, t_K)$ where $t_0 = 0 \leq t_1 \leq t_2 \leq \dots \leq t_K \leq 1$ with the interpretation that t_k is the time it takes to get the k 'th promotion. (Note that a professor can receive multiple promotions at the same time.) Denote by $\succsim$ the professor's (rational) preferences on the set of all possible careers. For any $\epsilon > 0$ and for any career t such that $t_K \leq 1 - \epsilon$, define $t + \epsilon$ to be the career $(t + \epsilon)_k = t_k + \epsilon$ for all k (i.e. all promotions are delayed by ϵ). The following are two properties of the professor's preferences:

Monotonicity: For any two careers t and s , if $t_k \leq s_k$ for all k then $t \succsim s$ and if $t_k < s_k$ for all k , then $t \succ s$.

Invariance: For every $\epsilon > 0$ and every two careers t and s for which $t + \epsilon$ and $s + \epsilon$ are well defined, $t \succsim s$ if and only if $t + \epsilon \succsim s + \epsilon$.

- a) (8) Formulate the set L of careers in which a professor receives all K promotions at the same time. Show that if $\succsim$ satisfies continuity and monotonicity, then for every career t there is exactly one career $s \in L$ such that $s \sim t$.

$$L = \{s | s = (\alpha, \dots, \alpha) \text{ for some } \alpha \in [0, 1]\}.$$

By monotonicity, $(0, \dots, 0) \succsim t \succsim (1, \dots, 1)$.

By continuity, there exists a bundle s on the interval connecting $(0, \dots, 0)$ and $(1, \dots, 1)$ such that $s \sim t$. Clearly, $s \in L$ since it is on the main diagonal.

By monotonicity, there is no other $s' \neq s$ in L such that $s' \sim t$.

- b) (6) Using the result from a), show by construction that if $\succsim$ satisfies continuity and monotonicity, then $\succsim$ has a utility representation.

Let $s^*(t) = (s(t), \dots, s(t))$ be the unique lottery in L that is indifferent to lottery t . Define the utility of t as $u(t) = -s(t)$.

Verify that $u(t)$ represents $\succsim$:

$$t \succsim v$$

$$\Leftrightarrow s^*(t) \succsim s^*(v) \text{ by transitivity}$$

$$\Leftrightarrow s(t) \leq s(v) \text{ by monotonicity}$$

$$\Leftrightarrow -s(t) \geq -s(v)$$

$$\Leftrightarrow u(t) \geq u(v) \text{ by definition of } u.$$

Hence, u represents the preferences.

- c) (10) Show that any preference which is represented by the function $U(t) = -\sum \alpha_k t_k$ (for some $\alpha_k > 0$ for each k) satisfies Monotonicity, Invariance and Continuity.

(Continuity can be argued with one sentence.)

MON: If $t_k \leq s_k$ for all k , then $-\Delta_k t_k \geq -\Delta_k s_k$ for all k . Consequently, $U(t) = \sum -\Delta_k t_k \geq \sum -\Delta_k s_k = U(s)$, and thus $t \succsim s$, and analogously in the strict case.

$$\begin{aligned} \text{INV: } t \succsim s &\Leftrightarrow -\sum \Delta_k t_k \geq -\sum \Delta_k s_k \Leftrightarrow -\sum \Delta_k t_k - \epsilon \sum \Delta_k \geq \\ &-\sum \Delta_k s_k - \epsilon \sum \Delta_k \Leftrightarrow -\sum \Delta_k (t_k + \epsilon) \geq -\sum \Delta_k (s_k + \epsilon) \Leftrightarrow t + \epsilon \succsim s + \epsilon. \end{aligned}$$

(4)

CON: U is continuous, so $\succsim$ is continuous.

- d) (6) One professor evaluates a career by the maximum length of time one has to wait for a promotion (i.e. the maximum $t_k - t_{k-1}$) and the smaller this number the better. Show that these preferences cannot be represented by the utility function described in (c).

By example. $(.2, .8) \sim (.1, .7)$, and thus preferences fail monotonicity.

2. (30 points) *Consumer choice* (after Rubinstein, B7)

Consider a consumer in a world of 2 commodities who has to make choices from budget sets parameterized by (p, w) , with the additional constraint that the consumption of good 1 is limited by some external bound $c \geq 0$. That is, in her world, a choice problem is a set of the form $B(p, w, c) = \{x | px \leq w \text{ and } x_1 \leq c\}$. The consumer has a demand function denoted by $x(p, w, c)$ which indicates her (unique) choice from $B(p, w, c)$.

Hint: Drawing diagrams might help you to understand the problem and finding the answers to the two following questions.

1. Assume that $px(p, w, c) = w$ and $x_1(p, w, c) = \min \left\{ 0.5 \frac{w}{p_1}, c \right\}$. Show that this behavior is consistent with the assumption that demand is derived from a maximization of the preference relation represented by the Cobb-Douglas utility function $u(x) = x_1^{0.5} x_2^{0.5}$. In other words, the behavior is rationalized by the preference relation represented by $u(x) = x_1^{0.5} x_2^{0.5}$.

(20 points)

If $0.5 \frac{w}{p_1} < c$, then $x(p, w, c) = (0.5 \frac{w}{p_1}, 0.5 \frac{w}{p_2})$ which is the unique maximizer of $u(x)$ on $B(p, w)$. Since $B(p, w, c) \subseteq B(p, w)$, this is also the unique maximizer on $B(p, w, c)$.

If $0.5 \frac{w}{p_1} > c$, then first note that the unique maximizer of $u(x)$ on $B(p, w)$ is not in $B(p, w, c)$. We show that $x(p, w, c) = (c, (w - p_1 c)/p_2)$ is the unique maximizer of $u(x)$ on $B(p, w, c)$.

Suppose not, and that there is $y \in B(p, w, c)$ such that $y \neq x(p, w, c)$ and $y \succsim x(p, w, c)$. It must be that $y_1 \leq c = x_1$. If $y_1 = x_1$, then $y_2 \leq x_2$ otherwise y would not be in $B(p, w, c)$ and y with $y_2 = x_2$ dominates any other such y because of the monotonicity of preferences. Thus $y_1 < x_1$ and by monotonicity $py = w$. Let $\bar{x} = (0.5 \frac{w}{p_1}, 0.5 \frac{w}{p_2})$.

This is the unique maximizer on $B(p, w) \supseteq B(p, w, c)$ and thus $\bar{x} \succsim y$. Now since $px(p, w, c) = w$, $x(p, w, c)$ can be written as a strict convex combination of y and $\bar{x}$ and Strict convexity by the preference relation then implies that $x(p, w, c) \succ y$. Thus $x(p, w, c)$ is the unique maximizer on $B(p, w, c)$.

2. Assume that $px(p, w, c) = w$ and $x_1(p, w, c) = \min \left\{ 0.5c, \frac{w}{p_1} \right\}$. Show that this consumer's behavior is inconsistent with preference maximization, that is, the behavior is not rationalizable.

Hint: Note that here, choice problems are defined by p, w, c , and remember that for rationalizability, the consumer's choices need to satisfy the weak axiom across all choice problems.

(10 points)

We show that this behavior violates the Weak Axiom. Consider (p, w, c) such that $.5c < w/p_1 < c$. Then $x(p, w, c) = (.5c, (w - p_1.5c)/p_2)$. Next consider (p, w, c') such that $.5c < .5c' < w/p_1 < c < c'$. Then $x(p, w, c') = (.5c', (w - p_1.5c')/p_2) \neq x(p, w, c)$. However, $B(p, w, c) = B(p, w, c')$ but then $x(p, w, c')$ and $x(p, w, c)$ cannot both be a unique preference maximizer.